

DropBox

Functional requirements:

user should be able to upload/download any file based on his preference,
user should be able to search for file based on his preference,
user should be able to sync up all the uploads

Non-functional requirements:

Data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem (Here eventual consistency is fine , but here we need 99.99% uptime
i.e Availability>>Consistency)

Core Entities:

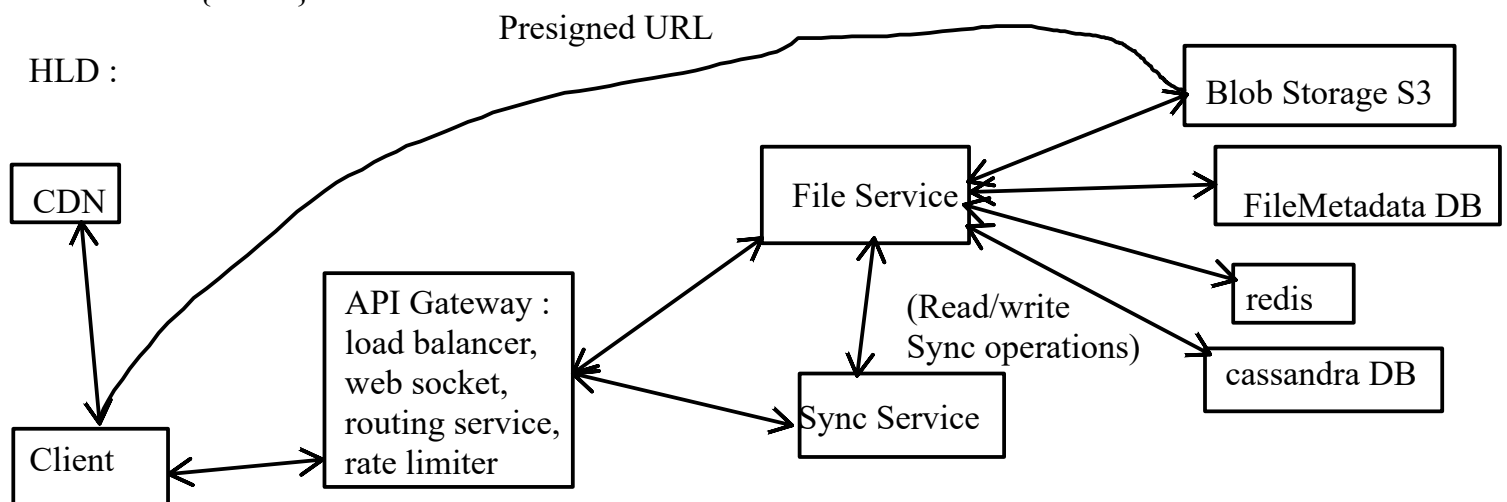
file,
user,
filemetadata

API :

POST:/uploads/file
{
 userID,
 {FileMetadata}
}

GET:/search/{fileID}

HLD :



- 1) we need to chunk 50GB file into multiple chunks
- 2) we need to fingerprint or hash each piece of these chunks separately
- 3) upload each piece of these chunks separately
- 4) update completed status for each piece of these chunks separately